

DSA0602 Data Handling and Visualaization - Slot A

Unit I - Class Activity 2

Date: 15.07.2026

S.No	Scenario	Task	Suggested Tool(s)
1	Smart City Air Quality Monitoring: A city has installed 150 air quality sensors across different locations. The sensors record PM2.5, NO ₂ , CO, temperature, and humidity every hour. Decision-makers struggle to identify pollution hotspots because of overlapping values.	Design multiple visualizations using Cartesian coordinates and different color scales to identify pollution hotspots. Compare sequential, diverging, and qualitative palettes and justify the most effective visualization.	Python, R, Tableau Public
2	Earthquake Risk Analysis: A geological survey contains earthquake magnitude (0.5–9.5), depth (1–700 km), latitude, longitude, and aftershock frequency. Small earthquakes dominate the dataset, making large events difficult to interpret.	Create visualizations using linear and nonlinear axes (log scale). Explain how logarithmic scaling changes the perception of earthquake distributions.	Python, R
3	Wind Direction Analysis for Airport Operations: An airport collects hourly wind speed and wind direction for one year. Pilots need an intuitive visualization showing prevailing wind directions and seasonal variations.	Develop a polar coordinate visualization (wind rose) to identify dominant wind directions and compare it with a Cartesian plot. Evaluate which visualization supports operational decision-making better.	Python, R
4	Hospital Patient Monitoring Dashboard: ICU patients have heart rate, oxygen saturation, blood pressure, respiration rate, and temperature recorded every minute. Doctors require a dashboard that instantly highlights abnormal conditions.	Build a dashboard using color as a highlighting tool. Select appropriate thresholds and explain why certain colors attract immediate attention without overwhelming users.	Python, R, Power BI
5	Global Climate Change Study: Researchers analyze temperature anomalies across 180 countries from 1980–2025. The objective is to show both positive and negative deviations from historical averages.	Design heatmaps using different diverging color palettes. Compare at least three palettes and justify which one communicates anomalies most effectively, including accessibility considerations.	Python, R
6	E-Commerce Sales Analytics: A retail company sells over 50 product categories across 30 regions. Management wants to distinguish categories while simultaneously representing sales volume.	Create visualizations that use color both to distinguish categories and represent numerical values. Discuss the challenges of combining qualitative and quantitative color encoding.	Python, Tableau Public
7	Satellite Image Vegetation Analysis: A satellite dataset provides NDVI values ranging from −1 to +1. Environmental scientists must identify healthy vegetation, barren land, and water bodies.	Apply different sequential and diverging color palettes to represent vegetation health. Evaluate which palette minimizes interpretation errors and improves readability.	Python, R
8	Cryptocurrency Market Volatility: Daily prices of Bitcoin, Ethereum, Solana, and Cardano span several years. Price values differ by several orders of magnitude, making direct comparison difficult.	Compare Cartesian plots with logarithmic axes. Use color strategically to distinguish cryptocurrencies while maintaining visual clarity. Explain the benefits and limitations of nonlinear axes.	Python, R
9	Marathon Runner Performance Analysis: A marathon organizer records runner speed, heart rate, distance covered, elevation, and running direction throughout the race. Coaches want to analyze pacing strategies and directional movement.	Develop both Cartesian and polar coordinate visualizations. Use color gradients to represent speed variations and recommend which visualization provides deeper performance insights.	Python, R
10	COVID-19 Vaccination Campaign Dashboard: Health officials monitor vaccination rates across districts, age groups, and vaccine types. They need visualizations suitable for both policymakers and the general public.	Design two visualization versions using different color palettes and highlighting strategies. Evaluate accessibility, interpretability, and effectiveness for audiences with color vision deficiencies.	Python, R, Tableau Public, Power BI